Economics and Business Administration — A.Y. 2025-26

Mock Exam — Financial Statement Analysis

Exercise 1)

The reclassified balance sheet of Società Alfa is as follows:

BALANCE SHEET				
	Year X-1	**Year X-1 %**	**Year X**	**Year X %**
ASSETS				
Cash and cash equivalents	225	*0.1%*	0	*0.0%*
Trade receivables	35,046	*20.1%*	68,750	*27.6%*
Inventories	25,225	*14.4%*	72,880	*29.2%*
Other current assets	450	*0.3%*	200	*0.1%*
CURRENT OPERATING ASSETS	**60,946**	***34.9%***	**141,830**	***56.8%***
Property, Plant & Equipment	108,187	*61.9%*	101,725	*40.8%*
Intangible assets	0	*0.0%*	0	*0.0%*
NON-CURRENT OPERATING ASSETS	**108,187**	***61.9%***	**101,725**	***40.8%***
NON-CORE ASSETS	**5,626**	***3.2%***	**5,988**	***2.4%***
TOTAL ASSETS	**174,759**	***100.0%***	**249,543**	***100.0%***
LIABILITIES & EQUITY				
Trade payables (goods and services)	30,926	*17.7%*	41,621	*16.7%*
Tax payables, other & TFR	28,947	*16.6%*	28,096	*11.3%*
CURRENT OPERATING LIABILITIES	**59,873**	***34.3%***	**69,717**	***27.9%***
Short-term bank debt	60,163	*34.4%*	127,850	*51.2%*
Medium/long-term bank debt	17,000	*9.7%*	11,592	*4.6%*
Other financial liabilities	978	*0.6%*	978	*0.4%*
FINANCIAL LIABILITIES (D)	**78,141**	***44.7%***	**140,420**	***56.3%***
Share capital & reserves	39,025	*22.3%*	36,745	*14.7%*
Net income for the period	-2,280	*-1.3%*	2,661	*1.1%*
EQUITY (E)	**36,745**	***21.0%***	**39,406**	***15.8%***
TOTAL LIABILITIES & EQUITY	**174,759**	***100.0%***	**249,543**	***100.0%***

INCOME STATEMENT				
	Year X-1	**Year X-1 %**	**Year X**	**Year X %**
Net Sales	**218,125**	***100.00%***	**335,550**	***100.00%***
Purchases of goods	-100,000	*-45.85%*	-150,000	*-44.70%*
Purchases of services	-53,802	*-24.67%*	-85,640	*-25.52%*
Change in inventories	-5,995	*-2.74%*	-7,985	*-2.38%*
Labour costs	-48,038	*-22.02%*	-66,740	*-19.89%*
Depreciation, amortization & write-downs	-10,412	*-4.77%*	-18,950	*-5.65%*
Core EBIT (EBIT_core)	**-122**	***-0.06%***	**6,235**	***1.86%***
Income from ancillary activities	3,110	*1.43%*	3,813	*1.14%*
EBIT (Operating Income)	**2,988**	***1.37%***	**10,048**	***2.99%***
Net Financial Expenses	-3,980	*-1.82%*	-6,675	*-1.99%*
EBT (Earnings Before Taxes)	**-992**	***-0.45%***	**3,373**	***1.01%***
Income Taxes	-1,288	*-0.59%*	-713	*-0.21%*
NET INCOME (NI)	**-2,280**	***-1.05%***	**2,661**	***0.79%***

Question 1

a) Calculate the following financial ratios:

	Year X-1	**Year X**
Operating NWC		
DSO (Days Sales Outstanding) — incl. VAT 20%		
DPO (Days Payable Outstanding) — incl. VAT 20%		
DIO (Days Inventory Outstanding)		
Working Capital Cycle		

b) Based on the ratios calculated and the financial statement data (note in particular the trend in net sales), comment on the causes behind the increase in Operating NWC. (max. 5 lines)

Question 2

a) Calculate ROE and decompose it into its drivers using the financial leverage formula (Modigliani-Miller):

	Year X-1	**Year X**
ROE		
CE		
RONA		
i = cost of debt		
Leverage ratio		
Tax Incidence		
FORMULA DECOMPOSITION		

b) Based on the ratios calculated, comment on the "levers" used by the company to increase ROE in Year X. (max. 5 lines)

Exercise 2)

The reclassified balance sheet of Società Commerciale Beta (Capital Employed approach) is as follows:

BALANCE SHEET		
	Year X-1	**Year X**
ASSETS		
Cash and cash equivalents	54	73
Trade receivables	4,003,082	8,860,000
Inventories	1,840,000	3,435,000
CURRENT OPERATING ASSETS	**5,843,136**	**12,295,073**
Property, Plant & Equipment	6,245,000	9,843,000
Intangible assets	24,500	36,000
NON-CURRENT OPERATING ASSETS	**6,269,500**	**9,879,000**
NON-Core ASSETS	**3,800,230**	**127,000**
TOTAL ASSETS	**15,912,866**	**22,301,073**
LIABILITIES & EQUITY		
Trade payables (goods and services)	4,112,236	5,629,150
Tax payables, other & TFR	205,830	214,775
CURRENT OPERATING LIABILITIES	**4,318,066**	**5,843,925**
Short-term bank debt	2,154,800	7,540,420
Medium/long-term bank debt	4,725,000	4,004,728
FINANCIAL LIABILITIES (D)	**6,879,800**	**11,545,148**
Share capital & reserves	4,500,000	4,500,000
Net income for the period	215,000	412,000
EQUITY (E)	**4,715,000**	**4,912,000**
TOTAL LIABILITIES & EQUITY	**15,912,866**	**22,301,073**

INCOME STATEMENT		
	Year X-1	**Year X**
Net Sales	**23,789,900**	**29,758,850**
Cost of goods sold	-16,724,000	-19,756,000
Purchases of services	-2,594,000	-4,308,000
Labour costs	-2,065,000	-3,875,312
Depreciation & amortization	-945,800	-1,819,455
Core EBIT (EBIT_core)	**179,728**	**1,738,538**
Income from ancillary activities	342,000	12
EBIT (Operating Income)	**521,728**	**1,738,550**
Net Financial Expenses	-229,443	-964,050
EBT (Earnings Before Taxes)	**292,285**	**774,500**
Income Taxes	-77,285	-362,500
NET INCOME (NI)	**215,000**	**412,000**

** Note: for simplicity, the result of non core activities is represented by the line item "Income from ancillary activities".*

a) Calculate ROI and decompose it into its drivers (core vs. non-core operations):

	Year X-1	**Year X**
ROI = EBIT / CE		
Core ROI		
Non-Core ROI		
Weight of Core Operations		
Weight of Non-Core Operations		
FORMULA DECOMPOSITION ROI		

b) What are the reasons behind the improvement in Core ROI? (max. 5 lines)

Exercise 3)

Classify each item using the Capital Employed reclassification scheme, marking with X if the item belongs to the category, or (X) if it is deducted from the category total.						
	Operating NWC	**Non-Current Operating Assets**	**Ancillary Assets**	**Financial Liabilities**	**Core Operating Costs**	**Equity**
Bond investments in portfolio						
Plant & equipment						
Surplus cash and cash equivalents						
Bond loans (issued)						
Finished goods inventories						
Current portion of mortgage loans						
Deferred tax provision						
Advance payments to suppliers for raw materials						
Legal reserve						
Post-employment benefit obligations (TFR)						
Trade payables						
Transportation costs						
Trademarks and patents						